Optimization Methods

Seminar 08

Constrained Optimality and Discrete Modelling: KKT, Duality, Integer Models

Theory handout

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This handout follows seminar 8. It goes from the first-order conditions of a constrained problem to the dual problem and the meaning of multipliers, and then turns to models with integer variables, where the same idea of a bound reappears as the relaxation. All nine problems of the class are solved in full.

Notation.
Problem: minimize $f(x)$ over $x \in \mathbb{R}^n$ subject to $g_i(x) \le 0$ and $h_j(x) = 0$. Multipliers $\lambda_i \ge 0$ for inequalities, ν_j of any sign for equalities. $L = f + \sum_i \lambda_i g_i + \sum_j \nu_j h_j$, $q = \inf_x L$, p^* and d^* the primal and dual optimal values. In the integer part a maximization keeps its own sign: there the relaxation bounds the optimum from above.

1 Optimality conditions with constraints

An unconstrained minimizer has $\nabla f = 0$. With constraints this fails as soon as the solution sits on the boundary: f would still like to go further, but the wall stops it. What remains true is a balance between the pull of f and the push of the walls that are touched.

Definition: Active set. For a feasible x, the indices i with $g_i(x) = 0$ form the active set $A(x)$. Constraints with $g_i(x) < 0$ are slack and do not restrict small moves.

Theorem: Karush–Kuhn–Tucker conditions. Let x^* be a local minimizer and suppose a constraint qualification holds there (for instance, the gradients of the active constraints and of the h_j are linearly independent, or the problem is convex and Slater's condition holds). Then there are λ^* and ν^* such that $\nabla_x L(x^*, \lambda^*, \nu^*) = 0$; x^* is feasible; $\lambda^* \ge 0$; and $\lambda_i^* g_i(x^*) = 0$ for every i.

Why. Stationarity says $-\nabla f(x^*) = \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \nu_j^* \nabla h_j(x^*)$: the direction in which f falls fastest points out of the feasible set, into the cone spanned by the outward normals of the touched walls. Any feasible move then has to go uphill.

Proposition: Sufficiency in the convex case. If f and every g_i are convex and every h_j is affine, a feasible point that satisfies the KKT conditions with some multipliers is a global minimizer.

Proof. With $\lambda^* \ge 0$ fixed, $x \mapsto L(x, \lambda^*, \nu^*)$ is convex and has zero gradient at x^*, so x^* minimizes it over all of $\mathbb{R}^n$. Complementary slackness and $h(x^*) = 0$ give $L(x^*, \lambda^*, \nu^*) = f(x^*)$. For a feasible x the added terms are ≤ 0, so $f(x^*) \le L(x, \lambda^*, \nu^*) \le f(x)$.

- Write L and the stationarity equations.
- Guess an active set A. Put $g_i = 0$ for $i \in A$ and $\lambda_i = 0$ for $i \notin A$, then solve the square system.
- Discard the candidate if some $g_i > 0$ or some $\lambda_i < 0$.
- In a convex problem the first survivor is the answer; otherwise compare the values of all survivors and check that a minimizer exists.

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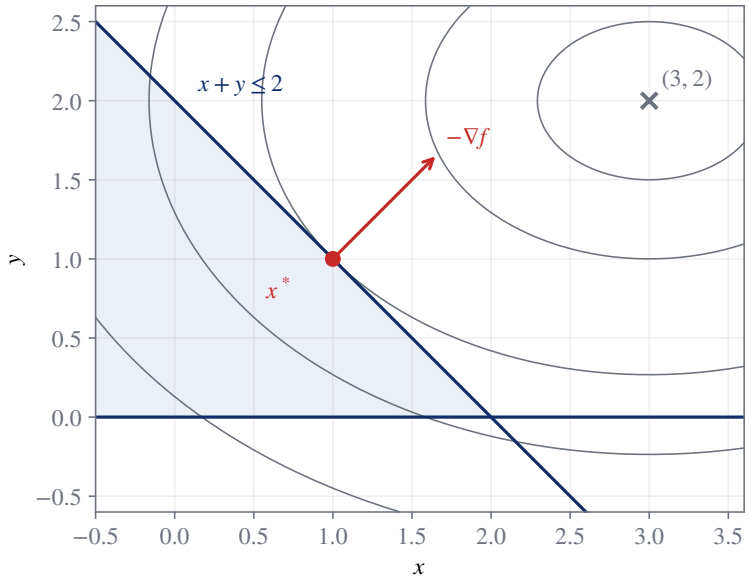


Figure. In A1 the minimizer lies on the line of g_1; $-\nabla f$ is a positive multiple of the normal of that line.

A1. Four guesses for the active set

On the board

Statement. Minimize $f(x, y) = (x-3)^2 + 2(y-2)^2$ subject to $g_1 = x + y - 2 \le 0$ and $g_2 = -y \le 0$. Go through all four active sets, find the KKT point and its multipliers, and give the first coordinate of x^*, λ_1^* and $f(x^*)$.

Solution.

1. No active constraint: the unconstrained minimizer $(3,\ 2)$ has $g_1 = 3 > 0$, so it is infeasible and at least one constraint binds.
2. Only g_2 active: $y = 0$, $\lambda_1 = 0$, and the first equation gives $x = 3$; the point $(3,\ 0)$ has $g_1 = 1 > 0$ and is rejected. Both active: $x + y = 2$ and $y = 0$ give $(2,\ 0)$, and stationarity gives $\lambda = (2,\ -6)$. A negative multiplier means that the floor $y \ge 0$ pushes the point the wrong way: rejected.
3. Only g_1 active, $\lambda_2 = 0$: $x = 3 - \lambda_1/2$, $y = 2 - \lambda_1/4$, and $x + y = 2$ gives $5 - 3\lambda_1/4 = 2$, so $\lambda_1 = 4$. The candidate is $x^* = (1,\ 1)$ with $\lambda^* = (4,\ 0)$; $y^* > 0$, so g_2 is indeed inactive.
4. At x^* the gradient is $\nabla f = (-4,\ -4) = -\lambda_1^* \nabla g_1$: minus the gradient points straight out through the line $x + y = 2$. The problem is convex, so the KKT point is the global minimizer, and $f(x^*) = 6$.

Answer. $x^* = (1,\ 1)$, $\lambda^* = (4,\ 0)$, $f(x^*) = 6$.

Trap. Active does not mean positive multiplier. The implication goes one way: $\lambda_i > 0$ forces $g_i = 0$, but $g_i = 0$ allows $\lambda_i = 0$. A constraint that just touches the unconstrained solution is the typical case.

A2. Three constraints

On your own

Statement. Minimize $f(x, y) = 3(x-5)^2 + (y-2)^2$ subject to $g_1 = x - 3.5 \le 0$, $g_2 = x + y - 5.5 \le 0$

and $g_3 = -y \le 0$. Find the KKT point, the active set and all three multipliers. Give λ_1^*, λ_2^* and $f(x^*)$.

Solution.

1. Without constraints f would sit at $(5, 2)$, but there $g_1 > 0$: the bound $x \le 3.5$ is violated, so g_1 must be tight. Put $x = 3.5$ and keep the other multipliers at zero for a moment. The y-equation $2(y-2) = 0$ gives $y = 2$, the x-equation $6(x-5) + \lambda_1 = 0$ gives $\lambda_1 = 9$.
2. At $x^* = (3.5,\ 2)$ the three constraint values are $g = (0,\ 0,\ -2)$. The sum constraint g_2 is tight as well, although nobody asked for it, while g_3 is slack, so $\lambda_3 = 0$ by complementary slackness. The active set is $\{1, 2\}$.
3. With both g_1 and g_2 in play, stationarity reads $6(x-5)+\lambda_1+\lambda_2 = 0$ and $2(y-2)+\lambda_2-\lambda_3 = 0$. At $y = 2$, $\lambda_3 = 0$ the second equation forces $\lambda_2 = 0$, and then the first gives $\lambda_1 = 9$ again. The gradients $(1, 0)$ and $(1, 1)$ of the tight constraints are independent, so these multipliers are the only ones: $\lambda^* = (9,\ 0,\ 0)$.
4. All four conditions hold, f is convex and the constraints are linear, so the point is the global minimizer with $f^* = 6.75$. The constraint g_2 touches x^* but carries no weight: delete it and nothing moves; delete g_1 and the answer changes.

Answer. $x^* = (3.5,\ 2)$, $\lambda_1 = 9$, $\lambda_2 = 0$, $\lambda_3 = 0$, $f^* = 6.75$.

A3. A linear objective on a disk

On your own

Statement. Minimize $x + 2y$ subject to $x^2 + y^2 \le 45$. Explain why the minimizer lies on the boundary, find it from the KKT conditions and check that it is global. Give y^*, λ^* and $f(x^*)$.

Solution.

1. With $c = (1,\ 2)$ and $g = x^2 + y^2 - 45$, stationarity is $c + 2\lambda(x, y) = 0$, so $(x, y) = -c/(2\lambda)$: the minimizer lies opposite to c, and $\lambda > 0$ is needed.
2. The circle is tight: $\|c\|^2/(4\lambda^2) = 45$ with $\|c\|^2 = 5$, so $\lambda^* = 0.1667$, $x^* = (-3,\ -6)$ and $f^* = -15$.
3. The objective is linear and the disk convex, so this KKT point is the global minimizer; the other point of the circle on the same line is the maximizer, with a negative multiplier.

Answer. $x^* = (-3,\ -6)$, $\lambda^* = 0.1667$, $f^* = -15$.

> **Check yourself.** In A1, why is the case with both constraints active rejected, although its point is feasible?
>
> **Answer.** Its multiplier for $y \ge 0$ comes out negative, $\lambda_2 = -6$: the wall $y = 0$ would have to pull the point instead of pushing it.

2 Lagrangian duality and prices

> **Definition: Dual function and dual problem.** $q(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$, possibly $-\infty$. The dual problem is to maximize q over $\lambda \ge 0$ and all ν; its value is d^*.

For fixed x the Lagrangian is affine in the multipliers, and an infimum of affine functions is concave. So the dual problem is always a concave maximization, whatever f and the constraints look like.

Theorem: Weak duality. For every feasible x, every $\lambda \geq 0$ and every ν, $q(\lambda, \nu) \leq f(x)$. Hence $d^* \leq p^*$.

Proof. $q(\lambda, \nu) \leq L(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x) \leq f(x)$, since $h(x) = 0$ and each $\lambda_i g_i(x) \leq 0$.

Theorem: Strong duality under Slater's condition. If the problem is convex with affine equalities, p^* is finite and some feasible point satisfies every inequality strictly, then $d^* = p^*$ and the dual maximum is attained.

Why. Read λ_i as a fine per unit of violation of $g_i \leq 0$. The dual asks for the fines at which the unconstrained minimizer of the fined objective is already feasible. Too low a fine and the minimizer cheats; the right fine is exactly the market price of the constraint.

Proposition: Multipliers measure sensitivity. Write the constraints as $g_i(x) \leq b_i$ and let $p^*(b)$ be the optimal value. Where p^* is differentiable and the optimal multipliers are unique, $\partial p^* / \partial b_i = -\lambda_i^*$.

Trap. Watch the direction of the inequality. In B1 the constraint is a lower bound $x + y \geq b$, written as $b - x - y \leq 0$; raising b tightens it, and the slope of p^* is $+\lambda^*$, not $-\lambda^*$.

B1. A dual function by hand

On the board

Statement. Minimize $x^2 + 3y^2$ subject to $x + y \geq 4$. Write the constraint as $g \leq 0$, find the dual function $q(\lambda)$, maximize it and compare d^* with p^*. Then treat the right-hand side b as a parameter and find dp^*/db at $b = 4$. Give λ^*, d^* and dp^*/db.

Solution.

1. $g = b - x - y \leq 0$ and $L = x^2 + 3y^2 + \lambda(b - x - y)$. From $2x - \lambda = 0$ and $6y - \lambda = 0$ the minimizer over (x, y) is $x = \lambda/2$, $y = \lambda/6$.
2. Substituting, $q(\lambda) = \lambda^2/4 + 3(\lambda/6)^2 + \lambda(4 - 2\lambda/3) = 4\lambda - \lambda^2/3$, a concave parabola. Its maximum over $\lambda \geq 0$ is where $q'(\lambda) = 4 - 2\lambda/3 = 0$: $\lambda^* = 6$ and $d^* = 12$.
3. The primal point comes from the same formulas, $(x^*, y^*) = (3,\ 1)$; it satisfies $x + y = 4$ and gives $p^* = 12 = d^*$. There is no gap, as Slater promises: the point $(4, 4)$ is strictly feasible.
4. For a general b the same steps give $\lambda^* = 3b/2$ and $p^*(b) = 0.75\, b^2$, so $dp^*/db = 6$ at $b = 4$, which is λ^*. The sign is plus because $x + y \geq b$ reads $-x - y \leq -b$: raising b tightens the constraint, and each unit costs λ^*.

Answer. $\lambda^* = 6$, $d^* = p^* = 12$, $dp^*/db = 6$.

B2. What one more unit of the resource is worth

On your own

Statement. Minimize $(x - 4)^2 + (y - 3)^2$ subject to $x + 2y \leq b$ with $b = 3$. Find $q(\lambda)$ and its

maximizer, then the primal solution. Predict p^* for $b + 0.5$ from the multiplier, then compute it exactly. Give λ^*, p^* and the exact p^* at the new b.

Solution.

1. With $c = (4,\ 3)$ and $a = (1,\ 2)$, $L = \|z - c\|^2 + \lambda(a^\top z - b)$ for $z = (x, y)$. Setting the gradient in z to zero gives $z(\lambda) = c - \frac{\lambda}{2}a$, that is $x = 4 - \lambda/2$, $y = 3 - \lambda$. The centre itself has $a^\top c$ far above $b = 3$, so the constraint will be tight.
2. Substituting $z(\lambda)$ gives the dual function $q(\lambda) = \lambda(a^\top c - b) - \frac{1}{4}\|a\|^2\lambda^2$, a downward parabola. Its top is at $\lambda^* = 2.8$ with value 9.8. The same λ put into $z(\lambda)$ gives $x^* = (2.6,\ 0.2)$, which satisfies $a^\top x^* = b$, and $p^* = \|x^* - c\|^2 = 9.8$: the two values agree, as Slater's condition promises (the origin is strictly feasible).
3. The constraint is an upper bound $a^\top z \le b$, so $dp^*/db = -\lambda^*$. For $\Delta b = 0.5$ the first-order forecast is $p^* - \lambda^*\Delta b = 8.4$.
4. Solving again with the larger right-hand side gives $\lambda = 2.6$, $x = (2.7,\ 0.4)$ and $p^* = 8.45$. The forecast misses by 0.05, a second-order effect: here $p^*(b) = (a^\top c - b)^2/\|a\|^2$ is a parabola in b, and its tangent lies below it.

Answer. $\lambda^* = 2.8$, $p^* = d^* = 9.8$; after the change of b, $p^* = 8.45$ (forecast 8.4).

B3. An equality and a free multiplier

On your own

Statement. Minimize $x^2 + 2y^2$ subject to $h = x + y - 6 = 0$. Find the dual function $q(\nu)$, maximize it over all $\nu \in \mathbb{R}$ and recover the primal point. Give ν^*, d^* and x^*.

Solution.

1. $L = x^2 + 2y^2 + \nu(x + y - 6)$. Minimizing over (x, y) gives $x = -0.5\nu$, $y = -0.25\nu$.
2. Substituting, $q(\nu) = -6\nu - 0.375\nu^2$, a downward parabola. Its top is at $\nu^* = -8$ with $d^* = 24$.
3. Then $x^* = (4,\ 2)$ lies on the line and $p^* = 24 = d^*$. The multiplier came out negative, which is allowed for an equality: it only says on which side of the line f would like to be.

Answer. $\nu^* = -8$, $d^* = p^* = 24$, $x^* = (4,\ 2)$.

> **Check yourself.** Can $q(\lambda)$ ever lie above p^*?
> **Answer.** No, for any $\lambda \ge 0$: that is weak duality, and it needs neither convexity nor Slater.

3 From a word problem to an integer model

Turning a story into a model takes three passes. Name the decisions and give each a variable, stating whether it is continuous, integer or binary. Translate every rule of the story into a constraint. Finally write the goal as a function of the variables. Most errors come from a rule that was never written down.

> **Definition: Integer program and its relaxation.** A mixed-integer linear program has a linear objective, linear constraints and some variables restricted to integers. Its LP relaxation is the same problem with the integrality requirement dropped.

Proposition: Dropping integrality gives a bound. For a maximization the relaxed optimum is at least the integer optimum; for a minimization it is at most. If the relaxed optimum happens to be integer, it solves the integer problem.

Proof. Every integer feasible point is feasible for the relaxation, so the relaxation optimizes over a larger set.

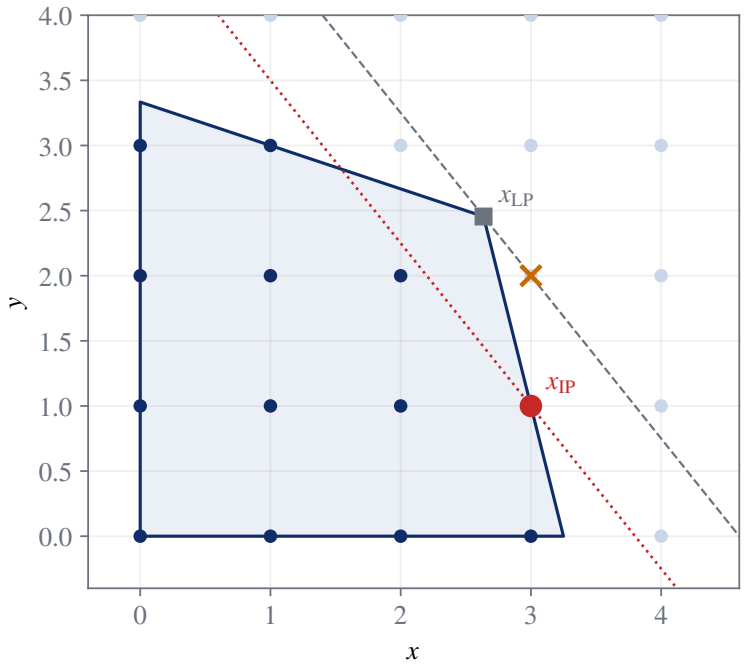


Figure. The relaxation of C1 peaks at a fractional vertex; the best integer point is not one of its roundings.

C1. Chairs and tables

On the board

Statement. A workshop makes chairs and tables. A chair brings a profit of 5, a table 4. A chair takes 1 hour of labour and 4 units of wood, a table 3 hours and 1 unit. There are 10 hours and 13 units of wood. Write the integer model, solve its LP relaxation, round the LP solution and find the integer optimum. Give z_{LP}, z_{IP} and the number of chairs in the integer optimum.

Solution.

1. Let x be the number of chairs and y the number of tables. Maximize $5x + 4y$ subject to $x + 3y \le 10$ (labour), $4x + y \le 13$ (wood), $x, y \ge 0$ and $x, y \in \mathbb{Z}$.
2. Drop integrality. At the vertices $(0, 0)$, $(3.25, 0)$, $\bar{x}$ and $(0, 3.3333)$ the objective is $(0,\ 16.25,\ 23,\ 13.3333)$; the best is the corner where both resources run out, $\bar{x} = (2.6364,\ 2.4545)$, with $z_{LP} = 23$.
3. Rounding down gives $(2,\ 2)$ with profit 18. The nearest lattice point $(3,\ 2)$ needs 14 units of wood, more than there is.
4. Checking the 13 feasible lattice points gives the integer optimum $(3,\ 1)$ with $z_{IP} = 19$: three chairs and one table, not the rounded plan. The relaxation promised 23, so the gap is 4, and no integer plan can beat the bound.

Answer. $z_{LP} = 23$ at $\bar{x} = (2.6364,\ 2.4545)$; $z_{IP} = 19$ at $(3,\ 1)$.

- **On/off with a fixed cost.** $x \le Mz$, $z \in \{0,1\}$, cost Fz: any output switches the cost on.
- **If A then B.** $z_A \le z_B$ for binary indicators.
- **At most k of n options.** $\sum_i z_i \le k$.
- **Either one constraint or the other.** $a^\top x \le b + M(1-z)$ and $c^\top x \le d + Mz$.
- **Product of binaries.** $w = z_1 z_2$ becomes $w \le z_1$, $w \le z_2$, $w \ge z_1 + z_2 - 1$.

Trap. Rounding the relaxed optimum is not a solution method. The rounded point may break a constraint, and even when it is feasible the true integer optimum can be several steps away, as in C1.

How big should M be

Any M at least as large as the largest value the variable can take is valid. Larger values keep the same integer solutions but weaken the relaxation, slow down branch and bound and, when M is huge, let a solver accept z a tolerance away from zero as switched off.

C2. Switch a machine on or not

On your own

Statement. An order of $D = 7$ units can be made on two machines. Machine i costs F_i to switch on and c_i per unit, and makes at most u_i units: $F = (20,\ 6)$, $c = (2,\ 5)$, $u = (5,\ 8)$. Write the model with binary z_i and $x_i \le M_i z_i$, and find the cheapest plan. Then relax $z_i \in \{0,1\}$ to $0 \le z_i \le 1$ and solve the relaxation for $M_i = u_i$ and for $M_i = 10\,u_i$. Give the three values.

Solution.

1. Decisions: output $x_i \ge 0$ and a switch $z_i \in \{0,1\}$ for each machine. Objective $\sum_i (F_i z_i + c_i x_i)$; demand $x_1 + x_2 = D$; capacity $x_i \le u_i$; link $x_i \le M_i z_i$, so that output forces the fixed cost. The smallest valid choice is $M_i = u_i$, and then the link already contains the capacity.
2. There are only four on/off patterns. Nothing on, or machine 1 alone with capacity $5 < D$, cannot cover the demand. Machine 2 alone costs $F_2 + c_2 D = 41$. Both on, with the cheap machine 1 at full capacity and the rest on machine 2, cost 46. The integer optimum is 41: one machine, the dearer one per unit.
3. Relax $z_i \in [0,1]$. Each unit on machine i now pays $c_i + F_i/M_i$. With $M = u$ these rates are $(6,\ 5.75)$, so all of the demand goes to machine 2 and the bound is 40.25, just below the integer optimum.
4. Take instead M_i equal to 10 capacities. The rates become $(2.4,\ 5.075)$: the fixed costs have almost vanished, machine 1 is filled to capacity and the bound drops to 22.15. The integer model is the same, but the relaxation now says little about it, and a branch-and-bound solver would have to work much harder.

Answer. Integer optimum 41 (machine 2 alone); relaxation 40.25 with $M = u$ and 22.15 with the inflated M.

C3. A knapsack and its relaxation

On your own

Statement. Four items have values $(8,\ 6,\ 5,\ 4)$ and weights $(3,\ 5,\ 2,\ 8)$; the knapsack holds 9. Write the model with binary x_i. Solve the LP relaxation by sorting the items by value per unit of weight, then find the best packing. Give the LP bound, the integer optimum and the number of items packed.

Solution.

1. Model: $z_i \in \{0,1\}$, maximize $\sum_i v_i z_i$ subject to $\sum_i w_i z_i \le W$ with $v = (8,\ 6,\ 5,\ 4)$, $w = (3,\ 5,\ 2,\ 8)$, $W = 9$. By the ratio v_i/w_i the order is $(1,\ 3,\ 2,\ 4)$.
2. The relaxation takes whole items in this order while they fit and a share 0.8 of the next one: the bound is 17.8.
3. Checking the subsets that fit, the best is the set $(1,\ 2)$ worth 14; the gap to the bound is 3.8. The greedy prefix is not necessarily the integer optimum.

Answer. Relaxation 17.8, integer optimum 14 with the items $(1,\ 2)$.

Check yourself. A binary variable z appears only in $x \le Mz$ and in the cost Fz with $F > 0$. What value does the relaxation give it?

Answer. The smallest allowed one, $z = x/M$: the relaxation pays only the share Fx/M of the fixed cost.